

— WORKING SESSION

Client-facing **AI-agents**

Four challenges of a customer-facing LLM assistant — and the control layer that makes its answers measurable, safe and auditable.

Market evidence

What 16 interviews across six jurisdictions say about client-facing GenAI — and what breaks inside a real assistant pipeline. 10xT Research · May–July 2026.

The state of client-facing GenAI in financial services

THE STUDY

- 16 conversations · 17 senior practitioners
- Tier-1 & digital banks · PSPs · vendors · AI & compliance consultants
- 6 jurisdiction clusters — EU · US · UK · Turkey · Eastern Europe · Central Asia
- Full maturity spectrum — from no AI deployed to a live voice assistant
- Discovery format · anonymized · cross-checked via consultants
- May – July 2026

SIX FINDINGS

01

Still the exception

Every retail institution runs or procures a chatbot — genuine GenAI toward the customer is live only at a handful.

02

Internal AI runs ahead

Incumbents pour AI into coding and back office while deliberately keeping it away from the customer for years.

03

Oversight doesn't scale — #1

Logs, sampling and manual review can't keep pace with volume; real-time control is missing — say both in-production tier-1s.

04

The only costed incidents

One bank runs a standing refund mechanism for wrong bot answers. Hallucination fear dictates architecture.

05

Regulation rising fastest

Driven by framework overlap and incoming AI laws — not enforcement: nobody could name a fine for an agent's answers.

06

Control plans deferred

Deployment plans are funded and dated; control plans are acknowledged and deferred. Leaders built oversight in-house.

There is rarely enough accurate, structured data to start from

LLM DISADVANTAGES

- Slow
- Expensive
- General
- Outdated
- Hallucinations

OUR PRACTICE

P1 We start with additional data collection and human annotation — training the intent classifier and populating the RAG knowledge base.

P2 We use an LLM for intent recognition.
Dimensionality reduction and distillation approaches — SLM.

Seven problem classes, ranked by real salience

1 Oversight scalability

Logging, sampling and manual review don't scale with dialogue volume; real-time control is missing. Named as the core problem by both in-production tier-1 banks.

2 Answer quality & truthfulness

The only recurring, costed incident pattern: standing customer refunds for wrong bot answers. Hallucination fear dictates architecture and stalls conversions.

3 Governance & accountability

Who owns the AI risk? Functions avoid personal exposure; months are lost to approvals. One global group suppressed customer-facing AI for ~2 years.

4 Regulatory compliance

Framework overlap plus incoming AI legislation — the only class with dated forcing events, and the fastest-rising concern.

5 Compliance content of communications

Mis-selling, unsuitable advice, misleading rates — harm without formal breach. Widely anticipated, rarely yet experienced.

6 Economics & technical maturity

Cost-per-answer vs quality, voice latency, vendor lock-in, legacy bots that cannot be converted.

7 Security

Prompt injection, jailbreaks, data leakage — ranked lowest by practitioners today; salience rises sharply once agents can take actions.

A chat interface raises every risk by a level

Users can put anything in a prompt, including “ignore previous instructions”. Even without malice, an LLM may respond with confidential or inaccurate information.

01

Prompt injection

Malicious instructions inserted into prompts to alter the response — e.g. revealing confidential information.

02

Jailbreaking

Bypassing built-in safety mechanisms to force the model to ignore its restrictions.

03

Prompt probing

Crafted queries that uncover hidden rules and internal constraints of the model.

04

Gray-box attacks

Adversarial prompts built on partial knowledge of model behaviour, aimed at its weak points.

05

Completion exploitation

Prompts designed so the model generates undesirable or harmful completions.

06

Biased prompt attacks

Prompts that trigger inherent biases, producing stereotypical or unfair responses.

Use separate LLM calls to keep dangerous input away from the main model — and to sanitize its output.

What actually breaks in a production assistant

Cases from 10xT's own end-to-end testing of a production-bound FS assistant, August 2026. Each component passed its own tests — the failures lived in the design.

APL-001 · DESIGN FLAW

Memory as an attack path

User messages were written to conversation history before any check ran. The answer node read that raw history — and executed attacks the parser and guard had already cut on the clean path.

Fix: history access removed from the answer node; the parser now rewrites follow-ups into standalone questions.

Memory is an input — validate everything the agent re-reads.

APL-002 · ENFORCEMENT

Advisory guard verdicts

A blocked query was still passed downstream “to be refused”. The model answered anyway — without RAG: the riskiest traffic got the least-controlled answers. One verdict per message also missed multi-question messages.

Fix: a splitter node breaks each message into questions; the guard issues per-question verdicts.

A verdict that relies on model cooperation is a suggestion, not a control.

APL-003 · DATA QUALITY

Consistent wrong grounding

One tag appeared in three documentation tables. Poor chunking meant the assistant answered from the wrong one — the same wrong answer, every run. The LLM-judge trusted the same chunks, so evaluation inherited the bug.

Fix: a reranker (at a latency cost) and a judge re-anchored to the source spec.

Stable errors point to the data pipeline, not the model.



10xT

— NEXT STEP

Thank you

Map these practices onto your assistant: metrics first, then the LLM-Guard control layer.

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